2 A soft rubber ball A of mass 0.50 kg is dropped from a height. At the same time another soft rubber ball B of mass 1.0 kg is projected vertically upward.

The two balls meet and collide head-on.

Fig. 2.1 shows the two balls just before the contact. Ball A is travelling downward at 2.0 m s ^{1} and ball B is travelling upward at 1.0 m s ^{1}.

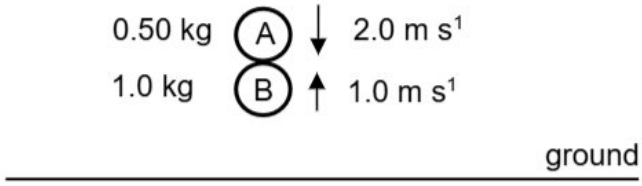


Fig. 2.1 (not to scale)

The two balls are in contact for 0.50 s.
The average contact force acting on ball A is 4.0 N.
Air resistance is assumed to be negligible.

(a) State and explain the magnitude and direction of the average contact force acting on B.

.....

.....

..... [2]

(b) The total impulse acting on ball A during contact is not equal in magnitude to the total impulse acting on ball B.

(i) State whether the total momentum of the system of two balls is conserved.

..... [1]

(ii) Determine the magnitude of the velocity of A at the instant the two ball separates.

magnitude of velocity of A = ... m s ^{1} [4]

[Total: 7]